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LLM-Enhanced Multi-Task Joint Learning Model for Misinformation Detection

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ABSTRACT

The coexistence of Human-Generated Content (HGC) and Artificial Intelligence-Generated Content (AIGC) versions of the same event on social media presents significant challenges for governmental governance and information regulation. In this study, we propose a Large Language Model-enhanced Multi-Task Joint Learning Model for Misinformation Detection (LMTMD) to address the challenge of mixed HGC and AIGC on social media. We design a two-stage instruction, leveraging large language models (LLMs) for data augmentation to generate AIGC versions of events. Furthermore, a novel unsupervised joint learning strategy is proposed, which incorporates content consistency contrastive learning and difference consistency learning. The strategy aims to preserve both the consistency of event content and the heterogeneity between AIGC and HGC. Extensive experiments conducted on real-world datasets, including Weibo and GossipCop, demonstrate that the proposed model outperforms state-of-the-art baselines, achieving a Consistent Match Accuracy (CM-Acc) of 77.21% on the Weibo dataset and 78.13% on the GossipCop dataset. Additionally, the model achieves AIGC detection accuracy rates of 90.58% on the Weibo dataset and 90.95% on the GossipCop dataset, thereby validating the effectiveness of both the model and the joint learning strategy. Our model can effectively adapt to the emerging scenario of mixed HGC and AIGC versions of events on social platforms and enriches the research perspective of misinformation detection.

1. Introduction

Social media platforms have emerged as a critical conduit for individuals to access information. However, the inherent freedom and complexity of these platforms make it challenging to comprehensively verify the information disseminated online. Numerous malicious users fabricate and disseminate misinformation on social media for various purposes. Misinformation can manifest in various forms, such as rumors, fake news, and satire, all of which can have detrimental effects on governmental governance and public safety (Meel & Vishwakarma, 2020; Kauk et al., 2025). The effective identification of misinformation on social media is both urgent and critical (Barve, et al., 2022). Discernible differences exist between genuine information and misinformation in terms of syntactic

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structures and lexical choices. The utilization of machine learning algorithms to extract these features has yielded promising results in misinformation detection (Shu et al., 2017). However, with the advancement of misinformation forgery techniques, relying solely on grammatical and stylistic features is insufficient for the effective detection of misinformation on social media platforms. Therefore, contextual information derived from social settings is increasingly considered supplementary for misinformation detection (Fang et al., 2024; Bazmi et al., 2023). For instance, Li et al. (2023) introduced propagation structure information and integrated it with graph contrastive learning for fake news detection. Song et al. (2019) proposed an early detection model for rumors by incorporating the temporal characteristics of event propagation.

However, the decline in public trust in media and the scarcity of relevant information during the initial stages of information dissemination significantly constrain the effectiveness of existing models that depend on social context information for the early detection of misinformation. Therefore, existing detection models must be capable of early detecting misinformation in the absence of social context information. Meanwhile, the advancement of large language models (LLMs) has enabled users to efficiently accomplish tasks such as manuscript writing, paper refinement, and text translation. LLMs have exhibited significant potential for diverse applications (Liu et al., 2024). The widespread adoption of LLMs has resulted in the coexistence of artificial intelligence-generated content (AIGC) and human-generated content (HGC) on social platforms. In particular, malicious users can readily utilize LLMs to camouflage misinformation, thereby evading existing detection methods (Wu et al., 2024). This scenario presents novel challenges for existing misinformation detection models. As illustrated in Fig. 1, when encountering HGC and AIGC versions of the same event, existing misinformation detection models generate contradictory judgments. Such inconsistency confuses users during information adoption decision-making processes and may even lead to erroneous decisions influenced by misinformation. To effectively adapt to the emerging scenario of mixed HGC and AIGC versions of the same event, existing models must be capable of providing consistent authenticity judgments across different versions of the event.

To address the aforementioned challenges and further investigate the impact of AIGC on misinformation detection, we integrate the AIGC detection task with the misinformation detection task, proposing a Large Language Model-enhanced Multi-Task Joint Learning Model for Misinformation Detection (LMTMD). To enhance the model's early detection capability, we focus on extracting textual information without relying on additional social context data. Specifically, we leverage the robust text representation capabilities of pre-trained language models to comprehensively extract semantic information from the text, fine-tuning the model to enhance its adaptability to the misinformation detection task. Furthermore, through dataset analysis (as illustrated in Fig. 2), we identify certain differences in the distribution of part-of-speech features between AIGC and HGC, as well as between misinformation and genuine information. Therefore, we incorporate manually engineered text features as auxiliary information, as detailed in Table 1. To effectively model the new scenario where HGC and AIGC versions of the same event coexist, we design a data augmentation method leveraging LLMs. Specifically, we develop a two-stage instruction: first, extracting the main content of the event from the original HGC version, and then expanding it using LLMs to generate an AIGC version, thereby creating sample pairs composed of HGC and AIGC versions of the event. In addition, to enable the model to consistently assess the authenticity of different versions of the same event, we propose a novel unsupervised joint learning strategy, comprising content consistency contrastive learning and difference consistency

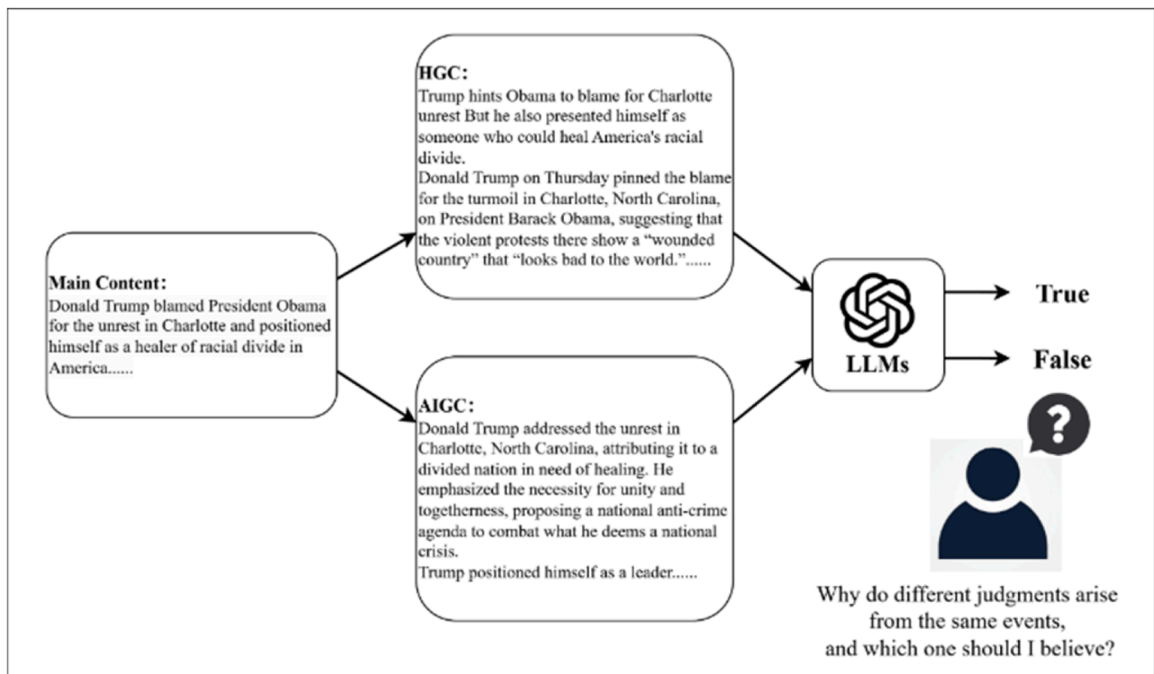


Fig. 1. The challenge of mixing AIGC and HGC.

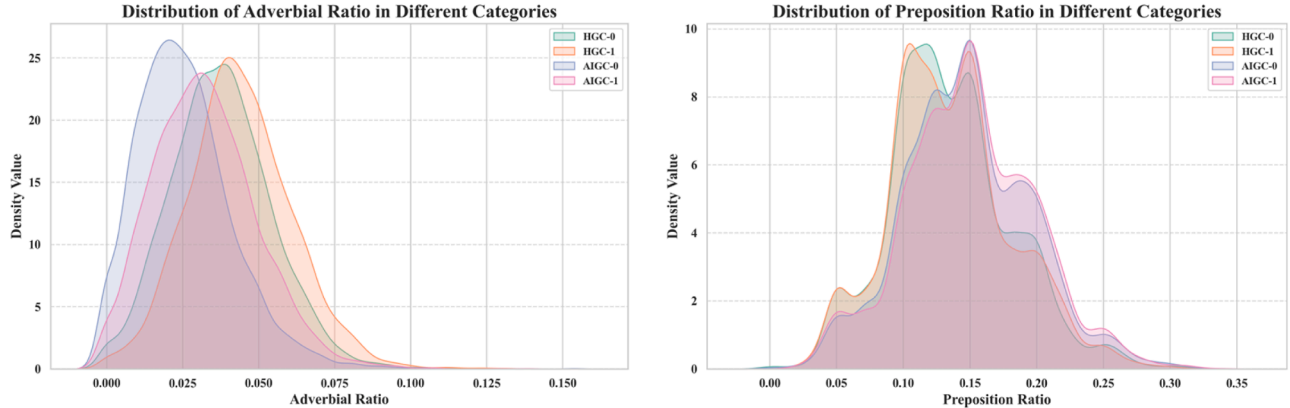


Fig. 2. Density Plot of Hand-crafted Feature Distribution.

Table 1
Hand-crafted Feature Description.

Category	Feature	Description
Feature Based on Part-of-Speech	Noun ratio	The ratio of nouns within a text.
	Preposition ratio	The ratio of prepositions within a text.
	Verb ratio	The ratio of verbs within a text.
	Adjectives ratio	The ratio of adjectives within a text.
	Adverbial ratio	The ratio of adverbs within a text.
Feature Based on Syntax.	Pronouns ratio	The ratio of pronouns within a text.
	Commonly used words ratio	The ratio of commonly words within a text.
	Stop words ratio	The ratio of stop words within a text.
	Syntax error	Presence of grammatical errors (1 indicates presence, 0 indicates absence).
Feature Based on Emotion.	Emotional polarity	The emotional tendency expressed in the text (1 denotes positive, 0 denotes neutral, and -1 denotes negative).

learning. The content consistency constraint between AIGC and HGC enables the model to prioritize content-based misinformation detection, reducing the likelihood of AIGC evading detection through structural or other non-content-related features. Simultaneously, the difference consistency learning between AIGC and HGC preserves the inherent distinctions between them, equipping the model with the capability to differentiate between AIGC and HGC. This unsupervised joint learning strategy enhances the learned feature representations, enabling their applicability to multi-task classification.

In summary, the coexistence of HGC and AIGC on social media platforms has introduced new complexities to the task of misinformation detection, highlighting the limitations of existing models. To better address these emerging challenges, this study aims to develop a robust model capable of effectively detecting misinformation in scenarios involving both HGC and AIGC. The goal of the research is to provide consistent and accurate authenticity judgments for both HGC and AIGC versions of the same event, thereby enriching the research landscape of misinformation detection in the era of LLM.

The innovations of this study are summarized as follows:

- (1) We propose a model named LMTMD for the detection of misinformation. Furthermore, a novel data augmentation method based on LLMs is designed, employing a two-stage instruction framework to reconstruct texts while preserving their core semantic content.
- (2) A novel unsupervised joint learning strategy is proposed, which integrates difference consistency learning between AIGC and HGC with content consistency contrastive learning. This strategy effectively retains the consistency of event content and the heterogeneity between AIGC and HGC, thereby enhancing the discriminative capability of the learned feature representations for multi-task classification.
- (3) Extensive experiments conducted on two real-world datasets augmented by LLMs demonstrate the superior performance of our proposed model and the effectiveness of the joint learning strategy.

2. Related works

2.1. Misinformation detection

2.1.1. Deep Learning-Based Misinformation Detection

The spread of misinformation can have devastating societal impacts, including incitement of violence and manipulation of public opinion (Kumar & Taylor, 2024). Consequently, an increasing number of researchers have focused on detecting misinformation on social media in recent years. Meanwhile, deep learning models have exhibited significant potential owing to their robust representation learning capabilities (Luo et al., 2023, Zhang et al., 2023). As a result, deep learning-based approaches for misinformation detection have garnered considerable attention. Xiao et al. (2024) proposed a novel text-syntax-based approach for fake news detection, employing a subgraph aggregation mechanism to capture subtle syntactic and semantic features within the text. Goldani et al. (2021) introduced a convolutional neural network incorporating margin loss and multiple embedding models for fake news detection. Alkhodair et al. (2020) designed a multi-objective recurrent neural network to enhance rumor identification. However, as misinformation forgery techniques advance, the distinction between meticulously crafted misinformation and genuine information is increasingly blurred (Yue et al., 2022). This has prompted researchers to integrate social context information to enhance detection performance (Bazmi et al., 2023). Xu et al. (2023) developed a multi-level text embedding approach utilizing graph neural networks, which incorporates the propagation structure of rumors to detect misinformation on social platforms. Bian et al. (2020) employed bidirectional graph convolutional neural networks to aggregate propagation and diffusion structure information, thereby enhancing the quality of rumor detection. In addition to social context information, features such as partisanship (Bazmi et al., 2023), user sentiment (Luvembe et al., 2023), and topic credibility (Chen et al., 2023) have also been utilized to optimize model performance.

2.1.2. Contrastive Learning-Based Misinformation Detection

Contrastive learning can acquire robust feature representations via self-supervised training (He et al., 2023). The key to contrastive

learning is to design appropriate contrastive loss functions that are specifically tailored to the research context (Wu et al., 2024). Commonly used measurement methods include InfoNCE loss (Ren et al., 2025), cosine similarity loss (Zhang et al., 2024) and Euclidean distance (Zheng et al., 2024) among others. Through the meticulous design of contrastive loss functions, contrastive learning has demonstrated significant advantages in the field of misinformation detection. For example, Ma et al. (2024) proposed a node-level contrastive learning framework, enabling the model to differentiate samples with analogous propagation structures. Zhang et al. (2025) integrated contrastive learning into the model pre-training stage to facilitate information fusion and derive robust feature representations for misinformation detection. Lu et al. (2025) designed a novel misinformation detection framework based on contrastive learning, which can achieve alignment and fusion across different fine-grained modal information while reducing the over-reliance on negative samples. To achieve the widespread application of contrastive learning, designing appropriate data augmentation strategies plays a crucial role (Ran & Jia, 2023). For example, Wang et al., (2023) proposed a sentence enhancement strategy based on a prompting mechanism, which enables text data augmentation while preserving more semantic information. Lin et al. (2024) started from the topological structure of error information propagation and combined it with a graph contrastive learning framework to design an adaptive sample augmentation strategy, which enhanced the model's performance in low-resource domains.

With the development of LLMs, a significant amount of AIGC has emerged on social media (Ayoobi et al., 2024), transforming the landscape of misinformation detection. Although the aforementioned models have achieved promising results, they have not adequately considered the complex implications of AIGC for misinformation detection tasks. Moreover, these models depend on contextual information, such as propagation structures and user data, for decision-making, which limits their ability to effectively assess the authenticity of posts from newly registered users. However, given that contrastive learning has demonstrated substantial potential in learning consistent features, this provides us with valuable technical references. In contrast to previous studies, we further consider the novel scenario of mixed AIGC and HGC on social platforms, as well as the emerging requirement for models to consistently evaluate the authenticity of both HGC and AIGC sharing the same content. Additionally, we focus on extracting discriminative features from textual content to enhance early misinformation detection and mitigate the adverse effects of its dissemination.

2.2. Applications of LLMs in Misinformation Detection

LLMs refer to natural language processing (NLP) models trained on extensive datasets and characterized by a large number of parameters, such as the GPT and LLaMA series. LLMs demonstrate robust capabilities in natural language understanding knowledge representation, advanced reasoning, and high-quality text generation (Yang et al., 2024). The formidable capabilities of LLMs mean that malicious actors can also exploit them to disseminate deceptive information and generate misinformation. It has become increasingly challenging for individuals to discern the authentic of such generated text (Boutadjine et al., 2025). This poses significant challenges for the identification and prevention of misinformation (Ayoobi et al., 2024). Simultaneously, the robust capabilities of LLMs also exert positive influences. The utilization of LLMs for misinformation detection has also demonstrated substantial potential. Qi et al. (2024) proposed a two-stage instruction tuning approach to unlock the extensive knowledge embedded in LLMs, thereby enhancing misinformation detection and improving the interpretability of model outcomes. Liu et al., (2024) designed instructions to extract specific forgery-related knowledge from LLMs and integrated this knowledge with textual and visual content for fake news identification. Despite their formidable capabilities in reasoning and generation, LLMs still exhibit certain gaps compared with meticulously designed or fine-tuned deep learning models in the domain of misinformation identification (Hu et al., 2024). Therefore, the integration of LLMs with other methods has emerged as a novel research direction. Teo et al. (2024) achieved enhanced performance in misinformation detection by integrating LLMs with machine learning techniques. Wang et al. (2025) enhanced the interpretability of misinformation detection by leveraging instruction engineering to unlock the powerful capabilities of LLMs within the framework of a Generative Adversarial Network (GAN). The collaboration between large models and small models has demonstrated substantial potential. Given the powerful zero-shot reasoning and text generation capabilities of LLMs, this study leverages them to extract part-of-speech tags and other handcrafted features from text, as well as to perform text data augmentation. Subsequently, we implemented the detection of misinformation by integrating a meticulously designed deep learning framework.

2.3. AIGC detection

AIGC refers to content generated by ChatGPT and other generative artificial intelligence (GAI) technologies (e.g., GAN networks, DM models), including text and images. With the rapid development and commercialization of GAI technologies, AIGC has raised a series of socio-ethical and privacy protection concerns. The identification of AIGC has gradually emerged as a new research direction, which can be primarily categorized into the detection of generated text and the detection of generated images.

Detection of Generated Text. Some models detect generated text by analyzing word frequency, complex phrase features, basic phrase features, and anomalous statistical features, employing statistical techniques for identification (Ippolito et al., 2020; Gehrmann et al., 2019). Schuster et al. (2020) found that language models tend to produce stylistically consistent text, a characteristic that facilitates the effective identification of generated text. Other researchers have focused on neural network-based detection methods (Rodríguez et al., 2022). Wang et al. (2023) designed a detection framework based on convolutional and self-attention networks to achieve detection of sentence-level generated text. Deep learning frameworks based on the Transformer architecture have also achieved certain progress in enhancing the robustness of generated text detection (Kuznetsov et al., 2024). Tian et al. (2024) proposed a multi-scale forward unsupervised training framework to address the challenge of detecting generated short texts without compromising the quality of long texts. Bhattacharjee et al. (2024) explored whether ChatGPT could effectively detect generated text. The detection of generated text remains an emerging field that warrants further investigation.

Detection of Generated Images. Current research primarily focuses on two aspects: extracting finer-grained image features and identifying unique traces left during the image generation process (Ciftci et al., 2020). Quan et al. (2018) designed and implemented a deep network with two cascaded convolutional layers at the bottom of the CNN to extract finer-grained features, introducing a local-to-global strategy to enhance the detection performance of existing generated images. Xi et al. (2023) optimized the recognition model framework by developing a robust dual-stream network composed of residual and content streams, capable of comprehensively capturing intrinsic anomalies generated by AIGC. Xu et al. (2023) constructed a hybrid neural network composed of an attention-guided feature extraction module and a vision transformer-based feature extraction module for detecting generated images.

Despite extensive exploration of AIGC detection, research on further assessing the authenticity of AIGC information remains limited. Therefore, our study focuses on the new challenges that the emergence of AIGC on social media poses to misinformation detection, attempting to bridge AIGC detection tasks with misinformation detection tasks.

3. Methodology

3.1. Problem definition

We are given a misinformation dataset $P = \{(p_{H1}, p_{A1}), (p_{H2}, p_{A2}), \dots, (p_{Hn}, p_{An})\}$, where n denotes the total number of sample pairs in the dataset, (p_{Hi}, p_{Ai}) represents the i -th sample pair, p_{Hi} is the HGC version of the event, and p_{Ai} is the AIGC version of the event. Meanwhile, p_{Hi} and p_{Ai} convey the same core content.

Each distinct sample pair is associated with a unique label value $y_i \in \{0, 1\}$, meaning that p_{Hi} and p_{Ai} share the same label y_i , where $y_i = 0$ indicates that the content of the sample pair is true, and $y_i = 1$ indicates that the information in the sample pair is false. For each sample, p_H and p_A are assigned unique labels $h_i \in \{0, 1\}$, where $h_i = 0$ denotes that the sample is human-written, and $h_i = 1$ denotes that the sample is AI-generated. The primary objective of this paper is to establish a mapping function $f(P) \rightarrow Y$, where Y represents the true label set. The main goal of this mapping function is to achieve consistent predictions of the authenticity of the textual information across different versions within each sample pair, i.e., $f(p_H) = f(p_A) = Y$.

3.2. Model

The overall framework of the proposed model, as illustrated in Fig. 3, comprises three main modules: the data enhancement module, the multi-task collaborative learning module, and the fusion classification module.

3.2.1. Data Enhancement Module

With the rapid development and widespread adoption of LLMs, their application scenarios have expanded significantly, and their usage has become increasingly convenient. LLMs can potentially be exploited to generate misinformation, resulting in a mixture of unverified HGC and unverified AIGC on social media platforms, thereby transforming the existing misinformation landscape (Guo, 2024). This complex scenario introduces new challenges for existing models. To effectively address this emerging challenge, we model

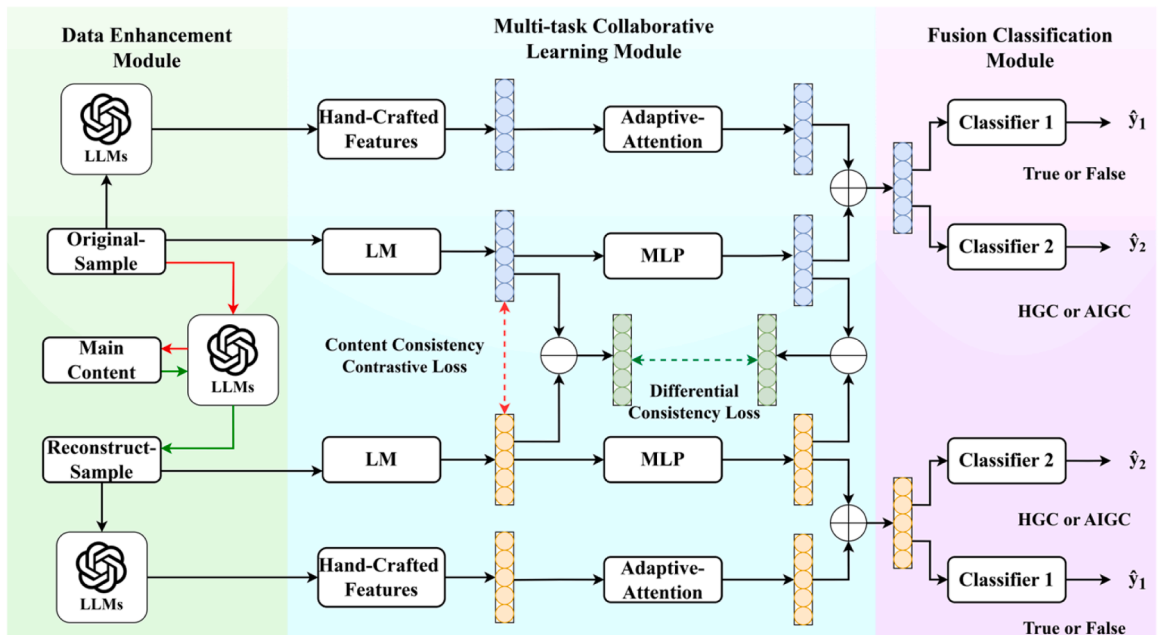


Fig. 3. Framework of LMTMD.

the coexistence of HGC and AIGC through a data augmentation approach. To ensure content consistency between AIGC and HGC, we design prompts, inspired by Wu et al., (2024), to guide LLMs in rewriting HGC while preserving content consistency, thereby generating reconstructed AIGC versions. To better model real-world scenarios and leverage the rich knowledge embedded in LLMs (Qi et al., 2024), we propose a two-stage prompting strategy for data enhancement. In the first stage, a prompt is utilized to extract the main content from the original HGC, as described below:

You are skilled at accurately extracting key information from text and highlighting critical points. Please summarize the main content of the provided {content} and calculates the word count.

In the second stage, instructions are utilized to guide the LLMs in generating a reconstructed AIGC version based on the extracted key information. To better simulate real-world conditions, no restrictions are imposed on the writing style of the AIGC. The instruction template is as follows:

You are skilled at generating diverse writing styles based on specific content. Please expands the {Main Content} while preserving its core message and targets a word count of approximately {word count}.

Finally, the reconstructed AIGC version is paired with the original HGC to form sample pairs(p_H, p_A). Given that the content of the restructured AIGC and the original HGC is consistent, they are assigned the same veracity label. Additionally, separate labels h are assigned to the restructured AIGC and the original HGC to indicate their respective origins (AIGC or HGC).

3.2.2. Multi-task Collaborative Learning Module

3.2.2.1. Hand-crafted feature extraction. Syntactic, grammatical, and emotional features of text provide rich supplementary information for misinformation detection. However, extracting these handcrafted features requires specialized knowledge and is both time-consuming and labor-intensive (Choudhary & Arora, 2021). To efficiently extract these handcrafted features, we leverage the strong zero-shot reasoning capabilities of LLMs (Wan et al., 2024) to mine features from text data in a batch-processing manner. Specifically, instructions are designed to guide the LLM to extract handcrafted features from each sample, as follows:

You are a linguistics expert specializing in text sentiment analysis. Extract the following features from the {content}: the proportion of nouns, verbs, prepositions, adjectives, adverbs, pronouns, common words, stop words, sentiment orientation, and the presence of grammatical errors. The sentiment orientation should be numerically encoded as follows: 1 for positive, 0 for neutral, and -1 for negative. Similarly, the presence of grammatical errors should be encoded as 1 (present) or 0 (absent). All results should be returned in a list format.

Using the aforementioned instructions, we extract the handcrafted features of p_H and p_A , denoted as z_H and z_A , respectively. The specific handcrafted features are summarized in Table 1. Since different handcrafted features may contribute unequally to the task, we dynamically integrate them using an adaptive attention mechanism. Specifically, we assign an initial weight a_i to each handcrafted feature and dynamically update these weights during model training to achieve adaptive feature fusion, as shown below:

$$z' = \sum_{i=0} a_i z_i \quad (1)$$

Here, z_i denotes the handcrafted feature in the i -th dimension, z' represents the updated handcrafted feature, and a_i denotes the trainable weight coefficient.

3.2.2.2. Text feature extraction. Pre-trained language models can generate dynamic text embeddings for downstream tasks. However, directly applying pre-trained language models may lead to domain bias issues (He et al., 2024). To address the specific requirements of misinformation detection tasks, we fine-tune the language model. Following Howard & Ruder (2018), we employ a parameter-freezing strategy to fine-tune the language model, thereby obtaining the initial text embedding representations. The detailed process is as follows:

$$x_H = LN(LM(p_H)) \quad (2)$$

$$x_A = LN(LM(p_A)) \quad (3)$$

Here, $x_H \in R^{b \times k}$ and $x_A \in R^{b \times k}$ denote the initial embedding representations of HGC and AIGC, respectively. b and k represent the batch size and embedding dimension, respectively. $LN()$ denotes the layer normalization operation, and $LM()$ refers to the pre-trained language model with partially frozen parameters.

Subsequently, the initial text embeddings are fed into a multi-layer perceptron (MLP) to generate the final text embeddings. The computation process is defined as follows:

$$x'_H = MLP(x_H), x'_A = MLP(x_A) \quad (4)$$

Here, x'_H and x'_A denote the final text embeddings of HGC and AIGC, respectively. $MLP()$ denotes a MLP comprising multiple linear layers and nonlinear activation layers. In this study, the MLP utilized for HGC and AIGC shares parameters.

Finally, the text embeddings and the handcrafted feature embeddings are concatenated to generate the final sample embedding representations X_H and X_A , which is utilized for the subsequent misinformation detection and AIGC identification.

$$X_H = x'_H \parallel z'_H, X_A = x'_A \parallel z'_A \quad (5)$$

Here, $\parallel$ denotes the concatenation operation

3.2.2.3. Joint learning strategy. To achieve consistent authenticity judgments for HGC and AIGC sharing the same content and enable accurate AIGC detection, we propose a joint learning strategy incorporating content consistency contrastive loss and difference consistency loss. Both x_H and x_A contain rich informational content. Inspired by Wu et al., (2024), we investigate consistent content representations in x_H and x_A from the perspective of content consistency to enhance misinformation detection. Specifically, we leverage the powerful self-supervised learning capability of contrastive learning (Wang et al., 2024) to derive content consistency representations for both HGC and AIGC. The content consistency contrastive loss is defined as follows:

$$L_{cc} = -\frac{1}{b} \ln \frac{\sum_{i=1}^b \exp(\theta(x_H^i, x_A^i) / \tau)}{\sum_{i \neq j} \exp(\theta(x_H^i, x_A^j) / \tau) + \sum_{i=1}^b \exp(\theta(x_H^i, x_A^i) / \tau)} \quad (6)$$

Here, x_H^i and x_A^i denote the embedding representations of HGC and AIGC in the i -th sample pair, respectively. In $\theta(x_H^i, x_A^i) = \text{sim}\left(\frac{x_H^i}{\|x_H^i\|_2}, \frac{x_A^i}{\|x_A^i\|_2}\right)$, $\text{sim}()$ represents the cosine similarity, τ is the temperature hyperparameter, and b denotes the batch size.

Given the inherent differences between AIGC and HGC, which facilitate their distinction, we propose a difference consistency loss to preserve robust differential representations for reliable AIGC detection. Specifically, the difference consistency loss is computed by first deriving the differential representations of the initial text embeddings diff1 .

$$\text{diff1} = x_H - x_A \quad (7)$$

Then, the differentiated representations (diff2) of the final text embeddings are computed.

$$\text{diff2} = x'_H - x'_A \quad (8)$$

Finally, we ensure the robustness of the differences between AIGC and HGC through a difference consistency loss, which is utilized for the task of AIGC detection.

$$L_{dc} = 1 - \frac{\text{diff1} \cdot \text{diff2}}{\|\text{diff1}\|_2 \|\text{diff2}\|_2} \quad (9)$$

Through the joint learning strategy integrating content consistency contrastive loss and difference consistency loss, the model learns content consistency representations while maintaining robust differential representations between AIGC and HGC. This learning strategy aims to endow the final text embeddings with the potential to perform multiple tasks. Furthermore, the proposed joint learning strategy is label-free and unsupervised, demonstrating strong transferability. The specific joint learning loss is as follows:

$$L_{mc} = a_3 L_{dc} + a_4 L_{cc} \quad (10)$$

Where a_3 and a_4 are adjustable hyperparameters.

3.2.3. Fusion Classification Module

The final sample embeddings are passed through distinct classifiers to perform different classification tasks, as described below:

$$\hat{y}_{hgc} = \text{classifier1}(X_H), \hat{y}_{aigc} = \text{classifier1}(X_A) \quad (11)$$

$$\hat{h} = \text{classifier2}(X) \quad (12)$$

Where, $\text{classifier1}()$ and $\text{classifier2}()$ denote the classifiers for misinformation detection and AIGC detection, respectively. $\hat{y}_{hgc}$ represents the predicted authenticity of HGC, while $\hat{y}_{aigc}$ represents the predicted authenticity of AIGC. X denotes the set of X_H and X_A , and $\hat{h}$ is the predicted value for the AIGC detection task.

The cross-entropy loss is employed as the objective loss function for both the misinformation detection and AIGC detection tasks. The formula is as follows:

$$L_{aigc} = -\sum_{i=1}^n h_i \log \hat{h}_i, L_{mid} = -\sum_{i=1}^n y_{aigc}^i \log \hat{y}_i - \sum_{i=1}^n y_{hgc}^i \log \hat{y}_i \quad (13)$$

Where, y_i denotes the ground-truth label indicating whether sample i is misinformation, and h_i denote the ground-truth label indicating whether sample i is AIGC. Following the approach of Li et al., (2023), the optimization objectives of the two tasks are summed to derive the final multi-task joint optimization loss. In addition, a joint learning loss is incorporated. Thus, the final loss function is defined as follows:

$$L = a_1 L_{mid} + a_2 L_{aigc} + L_{mc} \quad (14)$$

Where, a_1 and a_2 are adjustable hyperparameters. For ease of reference, Table 2 provides a detailed description of the primary notations used in this study, and the training process of our proposed method is outlined in Algorithm 1.

4. Experiments and discussion

4.1. Datasets

To validate the effectiveness of the proposed model, we evaluated its performance on two real-world benchmark datasets: the Weibo dataset (Ma et al., 2016) and the GossipCop dataset (Shu et al., 2020). The Weibo dataset is a Chinese dataset comprising verified news from the social media platform Weibo, whereas GossipCop is an English benchmark dataset widely adopted for misinformation detection tasks. The statistical details of the datasets are presented in Table 3. Herein, Max_len denotes the maximum text length, Min_len represents the minimum text length, and Avg_len indicates the average text length.

4.2. Experiment setting

In this study, the pre-trained model RoBERTa was employed as the text feature encoder. The proposed model was trained utilizing two NVIDIA 4070ti GPUs. The number of training epochs was set to 60. To mitigate overfitting, an early stopping mechanism with a patience period of 10 epochs was implemented. The dataset was randomly divided into training and testing sets with an 8:2 ratio. Five repeated experiments were conducted, and the average value was adopted as the final result. For the GossipCop dataset, the batch size was set to 8, while the hyperparameters a_1 , a_2 , a_3 , and a_4 were configured as 1, 0.5, 0.1, and 0.8, respectively. For the Weibo dataset, the batch size was set to 32, with the hyperparameters a_1 , a_2 , a_3 , and a_4 configured as 1, 0.3, 0.1, and 0.8, respectively. The initial text embedding dimension was set as 768, the learning rate was set to 1e-5, and the number of frozen layers in the pre-trained language model was fixed at 8.

In this paper, accuracy and F1 score were employed as evaluation metrics. Additionally, to address the research question, we proposed a novel evaluation metric, termed Consistent Match Accuracy (CM-Acc), which quantifies the consistency and accuracy in judging the authenticity of AIGC and HGC versions of the same event. This metric serves as the primary evaluation criterion in our study. The formula for calculating CM-Acc is defined as follows:

$$CM - Acc = \frac{\text{count}(\hat{y}_{hgc} = \hat{y}_{aigc} = y_i)}{n} \quad (15)$$

Where, $\text{count}()$ denotes the counting function used to determine the total number of sample pairs satisfying the condition, while n represents the total number of samples.

4.3. Baselines

To assess the performance of the proposed model, we compare it with the following baseline methods:

BERT (Devlin et al., 2019): A bidirectional pre-trained language model based on the Transformer architecture, which is trained

Table 2
Notations and Description.

Notation	Description
P	The set of all events
(p_H, p_{Ai})	Sample pairs consisting of HGC and AIGC content
p_H	Human-generated content
p_A	AI-generated content
z_H	Hand-crafted features extracted from HGC
z_A	Hand-crafted features extracted from AIGC
x_H	Initial text embeddings of HGC
x_A	Initial text embeddings of AIGC
x'_H	Final text embeddings of HGC
x'_A	Final text embeddings of AIGC
X_H	Embeddings of HGC samples
X_A	Embeddings of AIGC samples

Algorithm 1

Training Procedure for our Model.

Input: The dataset $P = \{(p_{H1}, p_{A1}), (p_{H2}, p_{A2}), \dots, (p_{Hn}, p_{An})\}$.
Output: The predicted probability of all samples.

```

1   for epoch in Epochs do
2     for each batch  $B$  in  $P$  do
3        $z'$  (handcrafted features of adaptive aggregation)  $\leftarrow$  Eq. (1)
4        $x_H$  and  $x_A$  (initialize text embeddings)  $\leftarrow$  Eq. (2) and Eq. (3)
5        $x'_H$  and  $x'_A$  (final text embeddings)  $\leftarrow$  Eq. (4)
6        $X_H$  and  $X_A$  (final sample embeddings)  $\leftarrow$  Eq. (5)
7        $\hat{y}_{hgc}$  and  $\hat{y}_{aigc}$  (predicted value of information authenticity)  $\leftarrow$  Eq. (11)
8        $\hat{h}$  (predicted value of AIGC)  $\leftarrow$  Eq. (12)
9       Assess the final loss by Eq. (14)
10      Update parameters of our model
11    end for
12  end for

```

Table 3

Datasets Statistics (the number of samples).

	Weibo (Chinese)		GossipCop (English)	
	HGC	AIGC	HGC	AIGC
Total	4,664	4,664	7,916	7,916
Fake	2,313	2,313	3,958	3,958
Real	2,351	2,351	3,958	3,958
Max_len	226	256	32,767	21,540
Min_len	2	7	239	114
Avg_len	104.68	121.42	3,547.40	2,163.65

using masked language modeling and next-sentence prediction tasks.

RoBERTa (Liu, 2019): An enhanced version of BERT that achieves more robust performance through increased data volume and optimized training strategies. RoBERTa has emerged as a important benchmark in the NLP field.

FNDNet (Kaliyar et al., 2020): A simple yet effective deep learning model for fake news detection, which utilizes a multilayer CNN architecture.

CNN-BiLSTM-AM (Xia et al., 2023): A hybrid model that integrates CNN, BiLSTM, and attention mechanisms.

BERT-LightGBM (Essa et al., 2023): A hybrid model that combines fine-tuned BERT with LightGBM for enhanced performance.

MSynFD (Xiao et al., 2024): A fake news detection method that employs a Transformer architecture with relative position awareness and leverages subgraph aggregation mechanisms to capture fine-grained syntactic relationships in text.

GLM-4 (GLM et al., 2024): A general-purpose pre-trained language model based on the Transformer architecture, which provides a robust foundation for various NLP tasks, such as text generation, machine translation, and sentiment analysis.

4.4. Results and analyses

Tables 4 - 7 summarize the performance of the proposed model and baseline models on the two datasets. Furthermore, paired t-tests were conducted to assess the statistical significance of the model results. Based on the results presented in Tables 4 and 5, the following conclusions can be drawn:

Table 4

Results of Misinformation Detection on the GossipCop Dataset.

	AIGC		HGC		CM-Acc
	Acc	F1-score	Acc	F1-score	
BERT	0.7580(0.0033)	0.7394(0.0056)	0.7648(0.0044)	0.7449(0.0061)	0.7117(0.0045)
RoBERTa	0.7797(0.0065)	0.7523(0.0041)	0.7828(0.0065)	0.7531(0.0051)	0.7222(0.0046)
FNDNet	0.7583(0.0076)	0.7387(0.0084)	0.7679(0.0066)	0.7468(0.0083)	0.7010(0.0082)
CNN-BiLSTM-AM	0.7714(0.0112)	0.7501(0.0081)	0.7754(0.0096)	0.7544(0.0094)	0.7068(0.0090)
BERT-LightGBM	0.7721(0.0107)	0.7654(0.0099)	0.7823(0.0109)	0.7668(0.0134)	0.7117(0.0091)
MSynFD	<u>0.7943</u> (0.0056)	<u>0.7765</u> (0.0046)	0.8233 (0.0073)	0.8045 (0.0077)	<u>0.7561</u> (0.0055)
GLM-4	0.7059	0.6902	0.8000	0.7536	0.6824
ours	0.8098 *(0.0033)	0.7878 *(0.0028)	<u>0.8151</u> (0.0055)	<u>0.7911</u> (0.0071)	0.7813 *** (0.0039)

Notes: * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$. For each metric, the best result is highlighted in bold, and the second-best result is underlined.

- (1) Compared to the authenticity detection of HGC, existing models face greater challenges in judging the authenticity of AIGC. Moreover, existing models tend to produce inconsistent authenticity judgments for AIGC and HGC with identical content, despite achieving high accuracy in single-type scenarios. The proposed model not only demonstrates strong performance in both HGC and AIGC single-type scenarios but also achieves more consistent authenticity judgments for AIGC and HGC with identical content in mixed scenarios. These results demonstrate the superior performance of the proposed model.
- (2) The experimental results indicate that the detection rates of misinformation in HGC and AIGC are closer on the GossipCop dataset, whereas a larger difference is observed on the Weibo dataset. Additionally, while the overall results on the GossipCop dataset are lower than those on the Weibo dataset, the detection rates of misinformation in AIGC remain relatively consistent across both datasets. Moreover, prior research has demonstrated that formal and standardized expression styles are more challenging to detect (Wu et al., 2024). Therefore, a potential explanation for this phenomenon is that AIGC, similar to written news, exhibits a formal style, whereas HGC on social platforms tends to adopt a more diverse and informal style. Consequently, on the GossipCop dataset, the detection rates of misinformation in HGC and AIGC are more similar, whereas on the Weibo dataset, HGC contains richer contextual information (e.g., language style, sentiment), leading to higher detection rates.
- (3) Fine-tuned language models outperform simpler deep learning models in terms of performance. This suggests that fine-tuning pre-trained language models can significantly improve their effectiveness in domain-specific tasks. Furthermore, our findings reveal that for text-based misinformation detection, emphasizing word-level semantic and syntactic information is particularly beneficial. For instance, the use of multi-head attention mechanisms and the construction of text syntactic graphs can more effectively capture word-level interactions, thereby enhancing the model's ability to identify misinformation.
- (4) Our results indicate that, compared to deep models specifically designed for misinformation detection tasks, LLMs do not exhibit a significant performance advantage. This implies that, despite the versatile capabilities of LLMs, their application in specific research domains may necessitate additional methodologies to optimize performance.

(5) Tables 6 and 7 present the results, which demonstrate that the method proposed in this study exhibits satisfactory performance in terms of precision and recall across both datasets. On the GossipCop dataset, our method achieved precision values of 0.8244 for AIGC and 0.8380 for HGC, outperforming all baseline models. Although the recall was slightly lower than that of MSynFD, the overall improvement in precision was more pronounced. On the Weibo dataset, our method demonstrated stronger generalization capabilities. It is noteworthy that the large language model GLM-4 exhibited a conservative characteristic of high precision but low recall in HGC detection. In contrast, our method achieved a better balance between the two metrics, thereby verifying its robustness across different content types and platform data.

The primary objective of this study is misinformation detection, with an additional focus on exploring the connection between AIGC detection and misinformation detection. As demonstrated in Table 8, our model achieves pleasant results in the AIGC detection task. This indicates that the representations learned by our proposed model are suitable for multi-task classification and further demonstrate that the integration of AIGC detection and misinformation detection tasks is beneficial.

4.5. Ablation study

To assess the effectiveness of the key components in our proposed model, we compare the full model with the following variants:

- handcrafted: This variant utilizes only textual information, excluding handcrafted features from the input samples. Furthermore, it removes the adaptive fusion module for handcrafted features implemented in the complete module.
- AIGC: This variant removes the AIGC detection task, retaining only the misinformation detection task during training.
- ldc: This variant eliminates the difference consistency loss during training, thereby simplifying the loss function structure.
- lcc: This variant removes the content consistency contrastive loss during training, ignoring the content correlation constraints between AIGC and HGC samples sharing consistent main content.
- lmc: This variant omits the joint learning strategy proposed in this study, utilizing only the cross-entropy loss as the optimization objective for the supervised classification task.

Table 5

Results of Misinformation Detection on the Weibo Dataset.

	AIGC		HGC		CM-Acc
	Acc	F1-score	Acc	F1-score	
BERT	0.7584(0.0073)	0.7566(0.0109)	0.8189(0.0065)	0.8154(0.0045)	0.6595(0.0079)
RoBERTa	0.7798(0.0079)	0.7705(0.0068)	0.8404(0.0072)	0.8366(0.0041)	0.6810(0.0082)
FNDNet	0.7302(0.0115)	0.7279(0.0111)	0.7950(0.0115)	0.7924(0.0115)	0.6247(0.0093)
CNN-BiLSTM-AM	0.7728(0.0127)	0.7651(0.0136)	0.8472(0.0079)	0.8417(0.0066)	0.6942(0.0107)
BERT-LightGBM	0.7644(0.0135)	0.7639(0.0120)	0.8321(0.0101)	0.8327(0.0106)	0.6782(0.0096)
MSynFD	<u>0.8138</u> (0.0077)	<u>0.8156</u> (0.0075)	<u>0.8709</u> (0.0069)	<u>0.8731</u> (0.0088)	<u>0.7488</u> (0.0049)
GLM-4	0.6660	0.6204	0.8148	0.7953	0.6188
ours	0.8248* (0.0050)	0.8234 (0.0073)	0.8812 (0.0057)	0.8913* (0.0068)	0.7721** (0.0044)

Notes: * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$. For each metric, the best result is highlighted in bold, and the second-best result is underlined.

Table 6

The Precision and Recall on the GossipCop Dataset.

	AIGC		HGC	
	Precision	Recall	Precision	Recall
BERT	0.7753	0.7067	0.7835	0.7099
RoBERTa	0.7843	0.7228	0.7943	0.7160
FNDNet	0.7902	0.6935	0.7876	0.7100
CNN-BiLSTM-AM	0.7715	0.7299	0.7735	0.7362
BERT-LightGBM	0.7933	0.7394	0.7949	0.7406
MSynFD	0.7991	0.7551	0.8255	0.7845
GLM-4	0.7474	0.5303	0.9630	0.6190
ours	0.8244	0.7543	0.8380	0.7492

Table 7

The Precision and Recall on the Weibo Dataset.

	AIGC		HGC	
	Precision	Recall	Precision	Recall
BERT	0.7766	0.7376	0.8421	0.7903
RoBERTa	0.7890	0.7528	0.8509	0.8228
FNDNet	0.7402	0.7160	0.7895	0.7953
CNN-BiLSTM-AM	0.7559	0.7745	0.8531	0.8306
BERT-LightGBM	0.7749	0.7532	0.8229	0.8427
MSynFD	0.8199	0.8113	0.8872	0.8594
GLM-4	0.7103	0.5508	0.8796	0.7257
ours	0.8046	0.8460	0.9110	0.8724

Table 8

Results of AIGC Detection.

	Acc	F1-score	Precision	Recall
GossipCop	0.9095	0.9100	0.9188	0.9014
Weibo	0.9058	0.9052	0.9201	0.8908

Figs. 4 and 5 illustrate the experimental results of our model and its variants on the GossipCop and Weibo datasets, respectively. Based on Figs. 4 and 5, the following conclusions can be drawn:

- (1) Compared to the complete model, the -handcrafted variant exhibited reductions in CW-Acc, AIGC-Acc, and HGC-Acc by 1.05%, 1.25%, and 1.27% on the GossipCop dataset, and by 4.11%, 3.63%, and 1.48% on the Weibo dataset, respectively. These results indicate that handcrafted features can offer supplementary information for assessing the authenticity of information. The performance degradation was more significant on the Weibo dataset compared to the GossipCop dataset. This may be due to the greater diversity in sample sources and forms within the Weibo dataset, as well as higher variability in wording and expression across samples. In contrast, AIGC content and news articles tend to exhibit more standardized wording and expression, leading to higher consistency. This implies that handcrafted features may have greater potential for application on complex social media platforms.
- (2) Compared to the complete model, the -AIGC variant exhibited reductions in CW-Acc, AIGC-Acc, and HGC-Acc by 2.69%, 2.07%, and 2.22%, respectively, on the GossipCop dataset, and by 2.61%, 1.77%, and 1.73%, respectively, on the Weibo dataset. These results highlight the effectiveness of integrating the AIGC detection task with the misinformation detection task in scenarios where AIGC and HGC coexist. By incorporating the AIGC detection task, the model learns the distinctions between AIGC and HGC during the training phase. This enables the model to leverage distinct features for different sample types when assessing information authenticity, thereby improving its adaptability.
- (3) Compared to the complete model, the -l_{dc} variant showed reductions in CW-Acc, AIGC-Acc, and HGC-Acc by 1.68%, 1.06%, and 1.59%, respectively, on the GossipCop dataset, and by 5.1%, 5.2%, and 1.95%, respectively, on the Weibo dataset. These results validate the effectiveness of the proposed difference consistency loss. By preserving robust differential features between AIGC and HGC, the proposed loss function enhances feature robustness, thereby improving the model's overall performance.
- (4) Compared to the complete model, the -l_{cc} variant led to reductions of 2.31%, 2.26%, and 1.96% in CW-Acc, AIGC-Acc, and HGC-Acc, respectively, on the GossipCop dataset, and reductions of 6.22%, 6.17%, and 1.36% in CW-Acc, AIGC-Acc, and HGC-Acc, respectively, on the Weibo dataset. These results underscore the effectiveness of the proposed content consistency contrastive loss. By focusing on content consistency between HGC and AIGC, the proposed loss function enables the model to

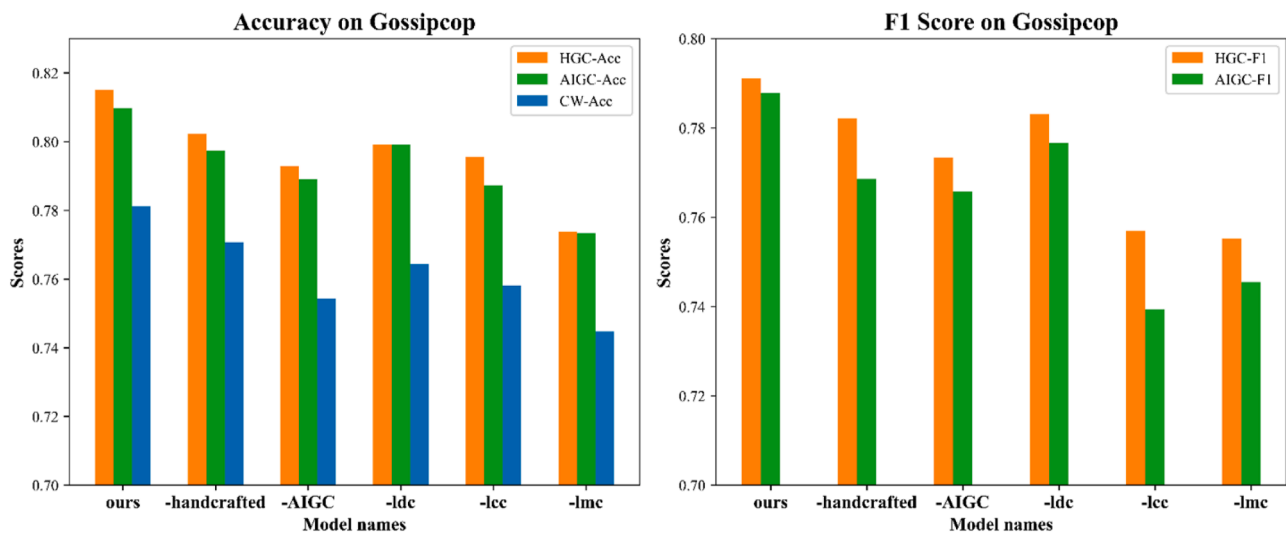


Fig. 4. Results of LMTMD variants on GossipCop dataset.

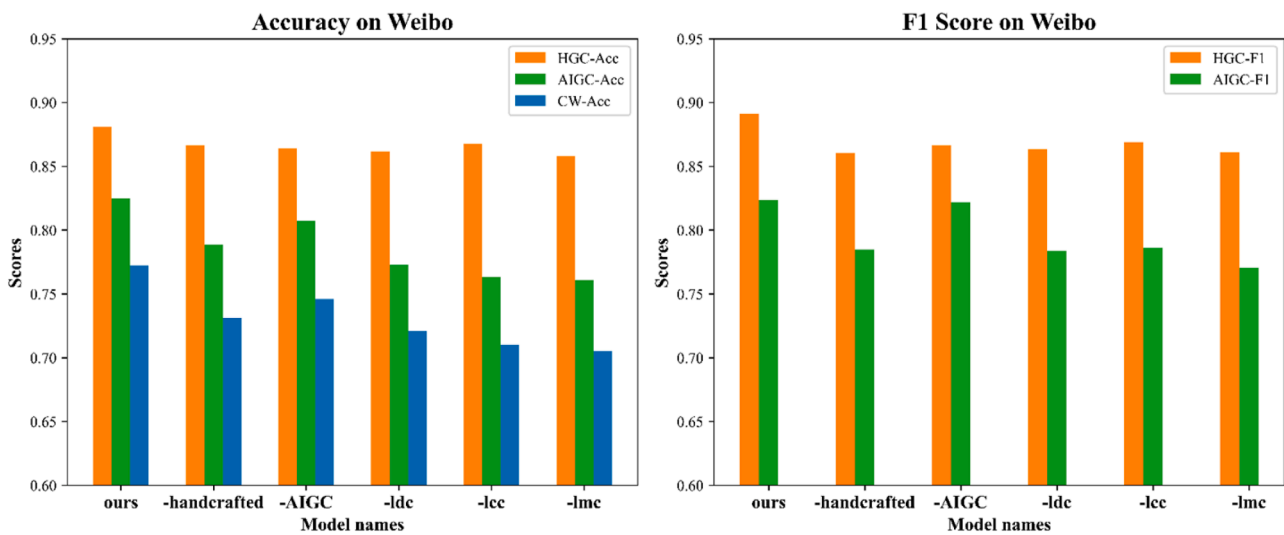


Fig. 5. Results of LMTMD variants on Weibo dataset.

more accurately assess the authenticity of AIGC and HGC information sharing the same main content, thereby improving the consistency of authenticity judgments for both content types.

- (5) Compared to the complete model, the -lmc variant exhibited reductions in CW-Acc, AIGC-Acc, and HGC-Acc by 3.65%, 3.65%, and 4.12%, respectively, on the GossipCop dataset, and by 6.68%, 6.38%, and 2.34%, respectively, on the Weibo dataset. These results confirm the effectiveness of the proposed joint learning strategy. By incorporating the unsupervised joint learning strategy, the model learns effective feature representations in a self-supervised manner from multiple perspectives. Furthermore, the joint learning loss function enables the model to fully leverage these features during training, thereby improving overall performance.
- (6) The proposed model outperformed all other variants on both the Weibo and GossipCop datasets, further demonstrating the effectiveness of each module. Additionally, the greater variation in metrics observed on the Weibo dataset suggests that AI-generated text may exhibit normative characteristics similar to news articles, highlighting a more pronounced discrepancy between AIGC and HGC on social media platforms.

4.6. Parametric analysis

Given that the primary objective of this study is misinformation detection, we investigated the impact of various hyperparameters on the model's performance in the misinformation detection task using the GossipCop dataset.

(1) In this study, the hyperparameter a_2 was adjusted from 0.2 to 0.7 to evaluate the impact of varying weights on model performance in the misinformation detection task following the integration of the AIGC detection task. As illustrated in Fig. 6, the model performance initially improved and subsequently declined as the weight increased. When $a_2=0.5$, the model achieved optimal performance across all evaluation metrics, suggesting that the integration of the AIGC detection task significantly enhances the model's misinformation detection capability.

Furthermore, as demonstrated in Fig. 7, superior performance in the AIGC detection task can be attained with lower values of a_2 , particularly on the Weibo dataset. This implies that AIGC on social media platforms may exhibit greater standardization in expression compared to HGC, making such standardized content more readily identifiable. Moreover, at lower values of a_2 , enhancements in AIGC detection performance are accompanied by improvements in misinformation detection performance, further highlighting the complementary relationship between the AIGC detection and misinformation detection tasks. However, when the weight exceeds an optimal threshold, the model tends to prioritize the AIGC detection task, consequently diminishing its performance in misinformation detection.

(2) To examine the influence of the weight a_3 of the difference consistency loss on model performance, a_3 was adjusted from 0.1 to 0.6. As depicted in Fig. 8, the CW-Acc metric demonstrated a declining trend as a_3 increased. Furthermore, it was observed that when $a_3>0.3$, the model's performance metrics for misinformation detection in both AIGC and HGC remained stable, with the F1 score even exhibiting an increase. This indicates that our model maintains high performance for misinformation detection in individual HGC and AIGC contexts. However, as a_3 increases, the model is more prone to generating inconsistent judgments for HGC and AIGC with identical content. A plausible explanation is that the feature-level differences between HGC and AIGC are significant, necessitating only minimal constraints. An excessively high a_3 value may lead the model to overemphasize the distinctions between AIGC and HGC, consequently impairing its overall misinformation detection performance.

(3) The value of a_4 was adjusted from 0.5 to 1 to evaluate the impact of the content consistency loss on model performance. The results are shown in Fig. 9. The model performance improved as the value of a_4 increased. However, when exceeded 0.8, the model exhibited fluctuations in the HGC-Acc metric, whereas the most critical metric, CW-Acc, demonstrated a declining trend. The optimal CW-Acc value was achieved at $a_4=0.8$, and therefore, was set to 0.8 in this study. The experimental results align with our theoretical expectations. Emphasizing content consistency enables the model to generate consistent authenticity judgments for HGC and AIGC with identical content. Furthermore, due to the significant differences between AIGC and HGC, capturing their consistent content necessitates greater model attention, which justifies the relatively high value of a_4 . However, when the weight of the content consistency loss is excessively high, the model may neglect other discriminative features for misinformation detection, thereby compromising the detection performance for individual content types and ultimately degrading the CW-Acc results.

(4) To investigate the influence of varying degrees of fine-tuning on the overall performance of pre-trained language models, the number of frozen language model layers was varied from 6 to 12. As shown in Fig. 10, the model's performance metrics exhibited fluctuations as the number of frozen layers varied. The optimal performance was attained when eight layers were frozen. Moreover, the parameter analysis results suggest that detecting misinformation in AIGC versions presents greater challenges.

4.7. Analysis of Contrastive Loss Function

To further investigate the effectiveness of the proposed joint learning strategy, we introduced an additional commonly used contrastive loss function based on Euclidean distance, in addition to the InfoNCE loss and cosine similarity loss that were originally employed. By pairwise combinations of these different loss functions to calculate L_{cc} and L_{dc} respectively. The detailed information is illustrated in Table 9.

The experimental results presented in Table 10 demonstrate the effectiveness of the proposed joint learning strategy. Among all the combinations, the contrastive learning strategy we employed exhibited the optimal performance, thereby validating the efficacy and rationality of our architectural design. Notably, the contrastive loss function based on Euclidean distance performed less satisfactorily. Of crucial importance is the observed performance degradation when the same loss function was applied to both objectives, which

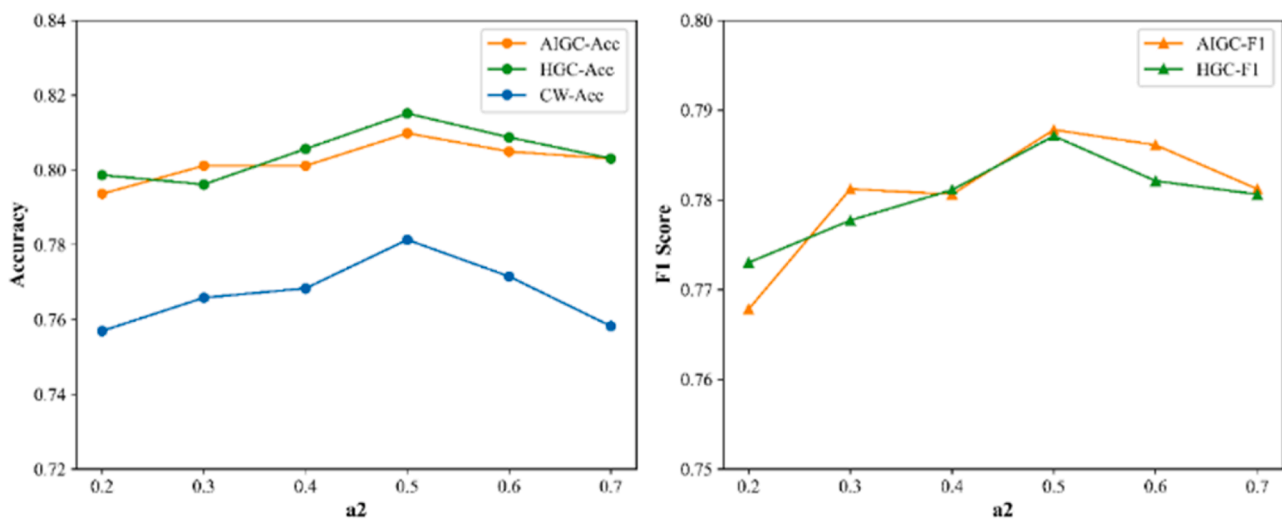


Fig. 6. Results of Different a_2 Values on Misinformation Detection Tasks.

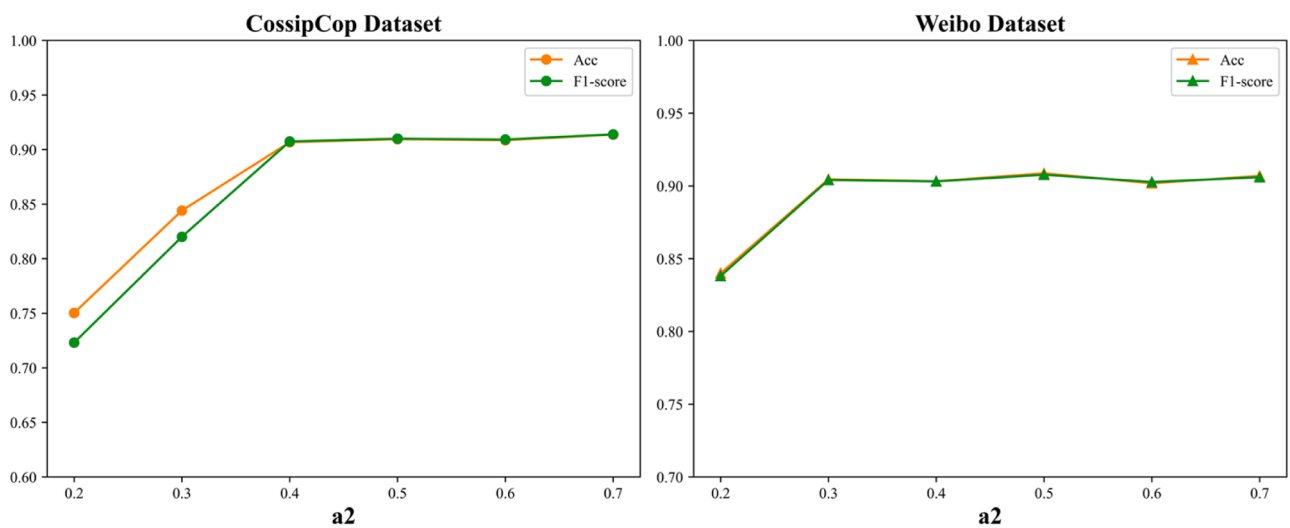


Fig. 7. The Impact of Different a_2 Values on AIGC Detection Tasks.

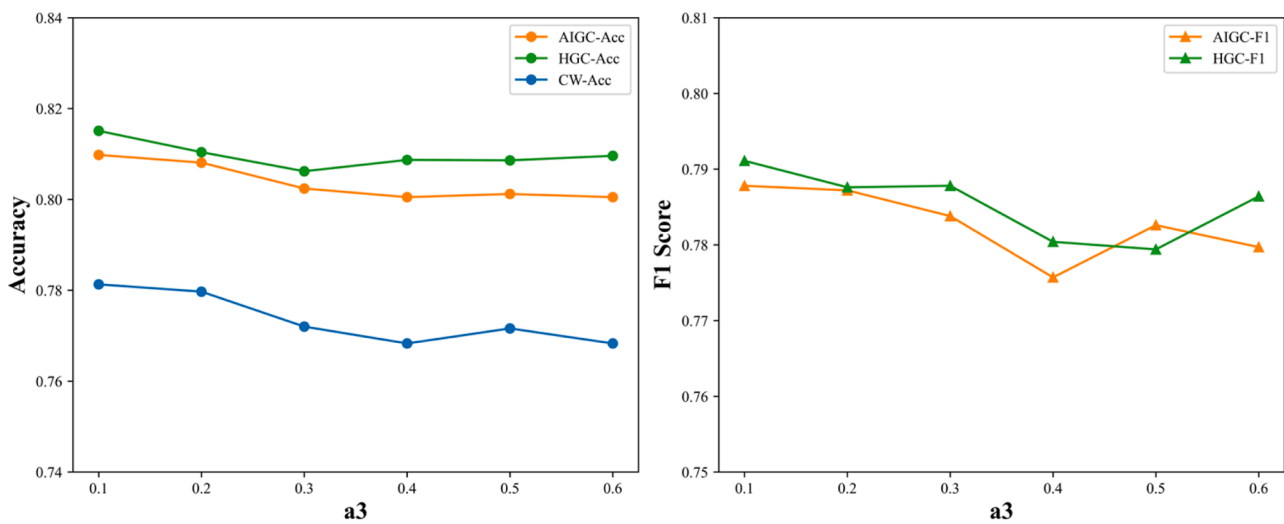


Fig. 8. Results of Different a_3 Values on Misinformation Detection Tasks.

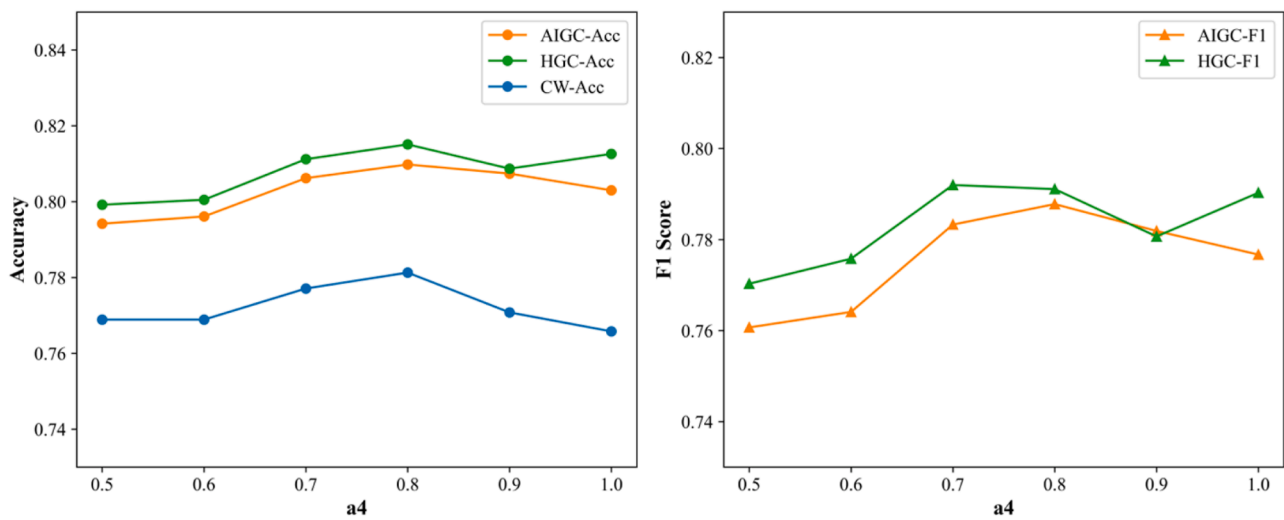


Fig. 9. Results of Different a_4 Values on Misinformation Detection Tasks.

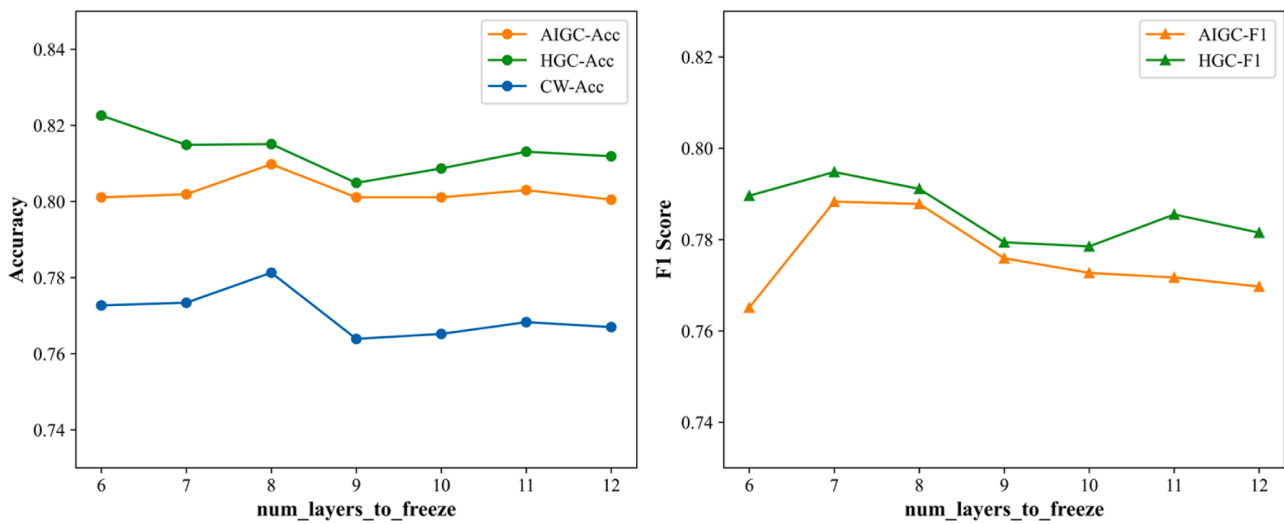


Fig. 10. Results of Varying Numbers of Frozen Layers.

Table 9
Summary of Contrastive Loss Functions.

Name	Calculation method	References method
InfoNCE loss	$-\frac{1}{b} \ln \frac{\sum_{i=1}^b \exp(\theta(x_H^i, x_A^i)/\tau)}{\sum_{i \neq j} \exp(\theta(x_H^i, x_A^j)/\tau) + \sum_{i=1}^b \exp(\theta(x_H^i, x_A^i)/\tau)}$	Ren et al. (2025)
Contrastive Loss Based on Euclidean Distance	$\frac{\text{dis}(x_i, x_j)}{\sum \text{dis}(x_i, x_j)}$	Zheng et al. (2024)
Cosine Similarity Loss	$1 - \text{cosine}(x_i, x_j)$	Zhang et al. (2024)

Notes: dis() denotes the Euclidean distance between vectors, while cosine() represents the cosine similarity between vectors.

Table 10
The CW-Acc on the GossipCop Dataset.

L_{cc} L_{dc}	InfoNCE loss	Contrastive Loss Based on Euclidean Distance	Cosine Similarity Loss
InfoNCE loss	0.7809	0.7734	0.7689
Contrastive Loss Based on Euclidean Distance	0.7771	0.7727	0.7680
Cosine Similarity Loss	0.7813(ours)	0.7740	0.7670

indicates the necessity for distinct similarity metrics to effectively model robust feature representations from different perspectives. Collectively, these findings underscore the significance of hybrid contrastive losses within a multi-task framework and suggest that different similarity measurement paradigms are complementary within a joint representation learning system.

4.8. Visual analysis

To gain deeper insights into the internal features learned by the model, t-SNE visualization was performed on the final sample embeddings (including both HGC and AIGC). It is worth noting that the t-SNE plot represents the dimensionality reduction results, so its axes have no specific labels. Fig. 11 and Fig. 12 illustrate that the model effectively distinguishes between true and false information across different sample types in the high-dimensional feature space. These findings validate the effectiveness and robustness of the proposed model. However, some boundary overlap was observed, suggesting that the model faces challenges in classifying certain samples. Furthermore, Fig. 13 and Fig. 14 demonstrate that the proposed model effectively clusters HGC and AIGC within true and false information in the high-dimensional feature space, facilitating consistent authenticity identification of HGC and AIGC. Nevertheless, some sample dispersion was observed, indicating potential areas for further improvement in the proposed model.

As depicted in Fig. 15, the importance of different handcrafted features varies significantly across categories in misinformation detection tasks. For HGC, syntax errors, stop words ratio, and the pronouns ratio are assigned relatively higher weights. In contrast, for AIGC, emotional polarity, the nouns ratio, and the pronouns ratio are assigned higher weights. Notably, emotional polarity and pronoun ratio are critical features in both categories, suggesting that handcrafted features can offer valuable supplementary information for misinformation detection tasks. Moreover, the informational contribution of different handcrafted features varies significantly.

5. Conclusion

The coexistence of HGC and AIGC versions of the same event on social media has substantially heightened the complexity of misinformation detection. Given this emerging social phenomenon, maintaining social media credibility, ensuring information authenticity on platforms, and achieving early detection and governance of misinformation have become pressing societal challenges. To address this challenge, we proposed a model named LMTMD. The strong zero-shot reasoning capabilities of LLMs were leveraged through instruction design to model scenarios involving mixed HGC and AIGC versions of the same event. Additionally, a novel unsupervised joint learning strategy was designed to enhance the model's ability to consistently assess the authenticity of both HGC and AIGC versions of the same event, enabling the learned representations to support multi-task classification. Extensive experiments conducted on real-world public datasets demonstrate that the proposed model outperforms baseline models in misinformation detection while achieving pleasant results in AIGC detection tasks.

The study yields the following key findings: (1) Compared to HGC, determining the authenticity of AIGC is more challenging. (2) Text generated by LLMs without specific fine-tuning may exhibit normativity comparable to professionally formulated news content. (3) The integration of the AIGC detection task enhances the performance of misinformation detection, demonstrating a complementary relationship between the two tasks.

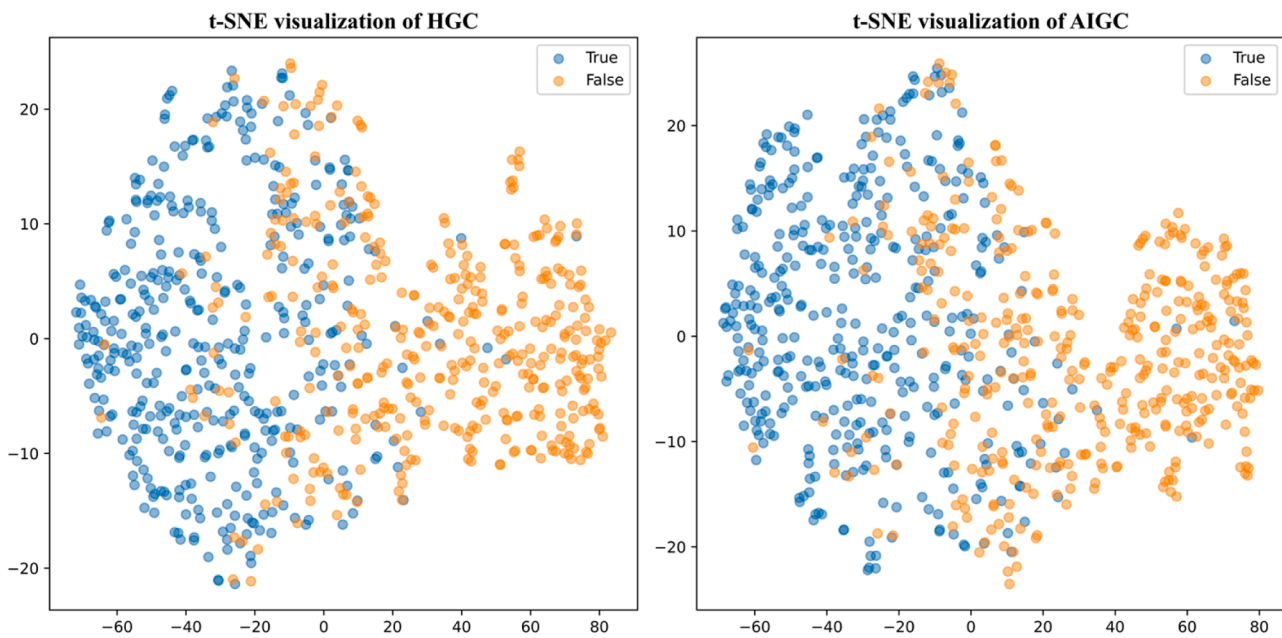


Fig. 11. t-SNE Visualization of HGC and AIGC on the GossipCop Dataset.

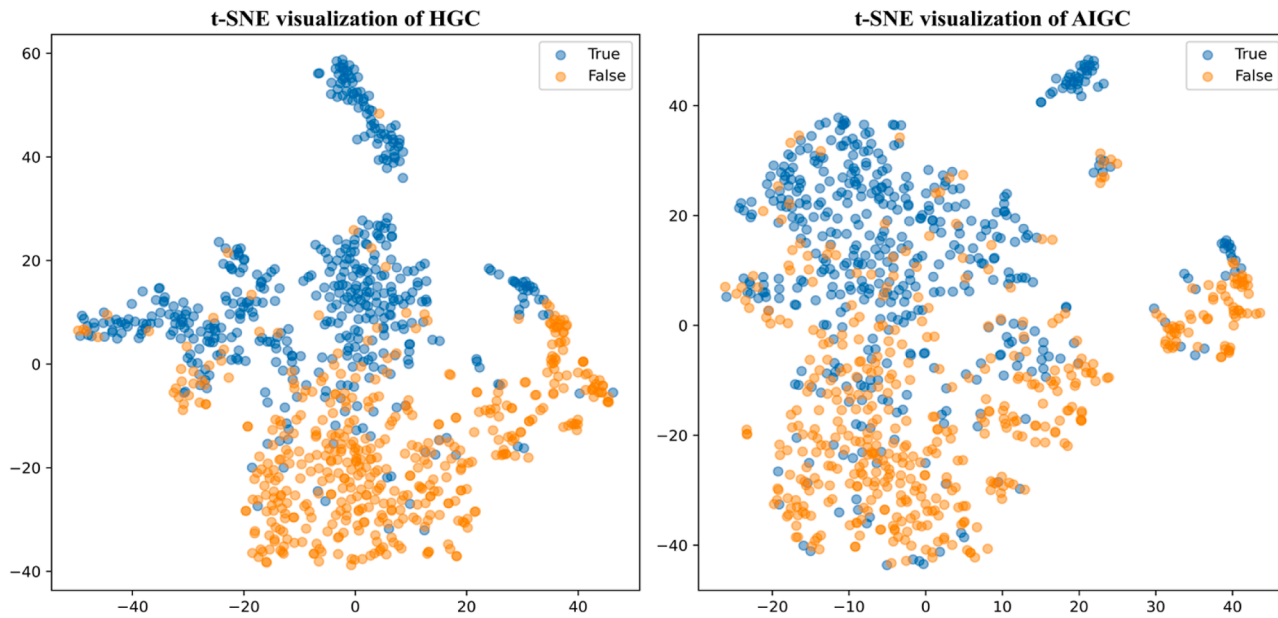


Fig. 12. t-SNE Visualization of HGC and AIGC on the Weibo Dataset.

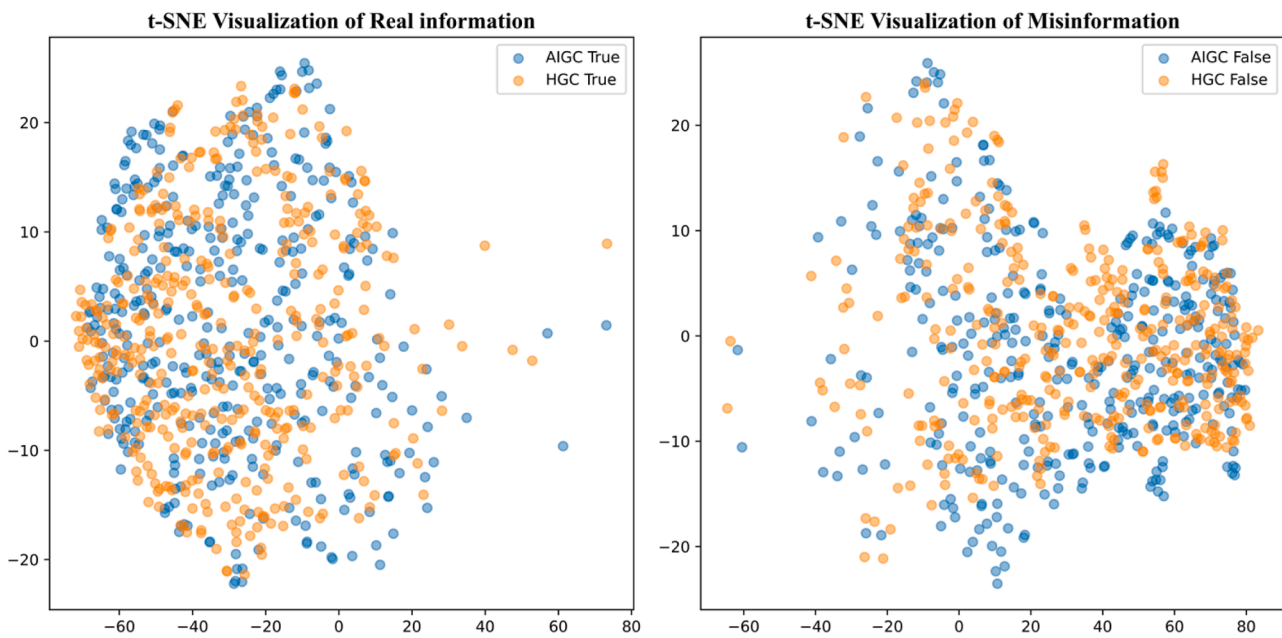


Fig. 13. t-SNE Visualization of Authentic and Misinformation on the GossipCop Dataset.

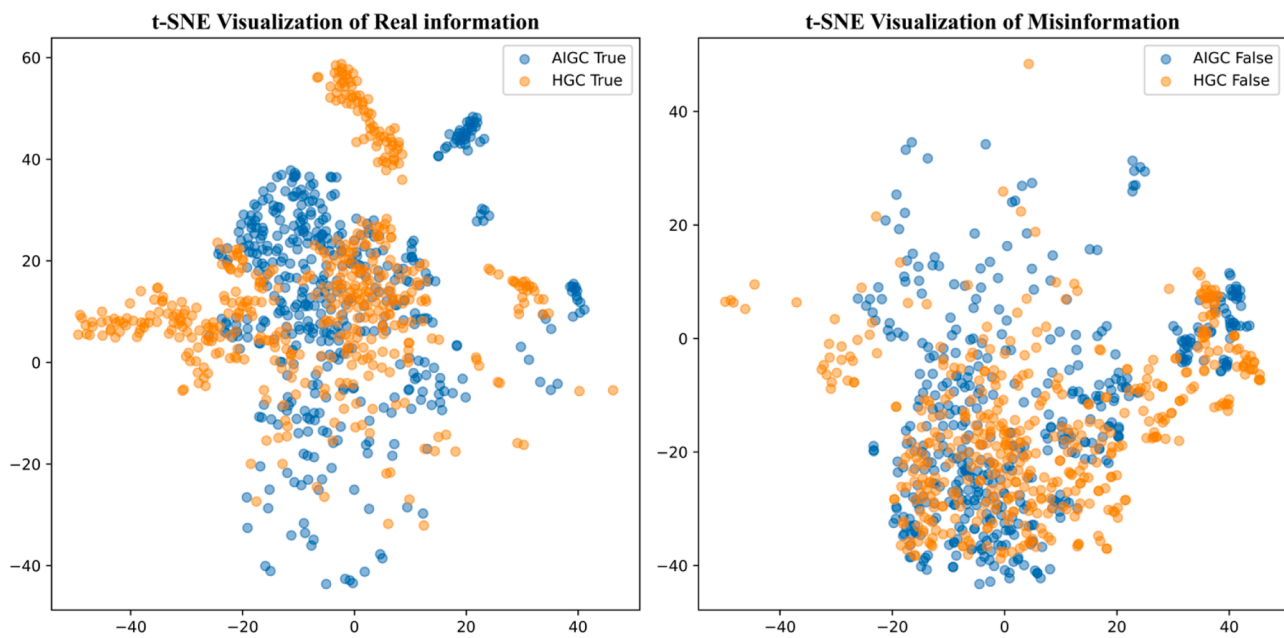


Fig. 14. t-SNE Visualization of Authentic and Misinformation on the Weibo Dataset.

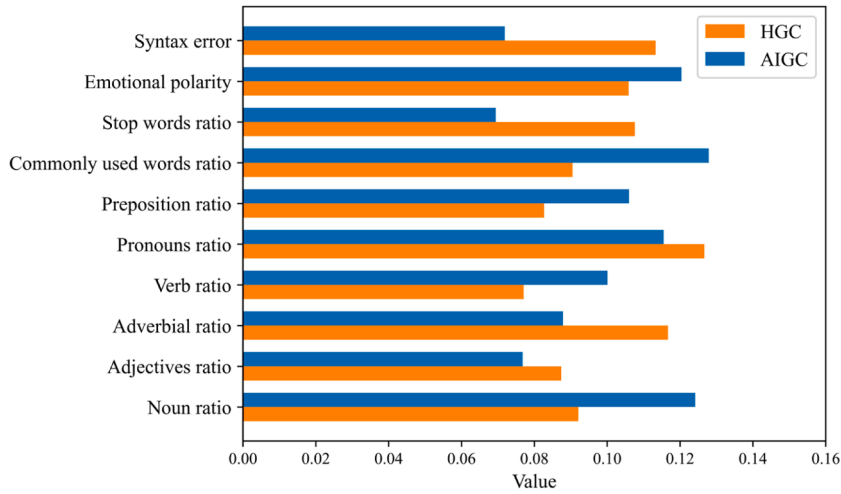


Fig. 15. Visualization of Hand-crafted Feature Weights.

6. Discussion

6.1. Theoretical implications

This study advances the novel understanding of misinformation detection in the context of the coexistence of HGC and AIGC on social media platforms. By proposing a model named LMTMD, this research integrates AIGC detection with misinformation detection, thereby enriching the theoretical framework and theoretical background of content authenticity assessment. The novel unsupervised joint learning strategy, which incorporates content consistency contrastive learning and difference consistency learning, provides a robust methodological approach to preserving both the consistency of event content and the heterogeneity between AIGC and HGC. This dual focus not only enhances the model's ability to detect misinformation but also contributes to the broader field of NLP by demonstrating the effectiveness of joint learning strategies in addressing complex, mixed-content scenarios. Additionally, the study underscores the significance of leveraging LLMs for data augmentation and feature extraction. While acknowledging the potential risks associated with LLMs, the research also recognizes their transformative positive impacts.

6.2. Practical implications

The practical significance of this study lies in its potential to improve the accuracy and reliability of misinformation detection on social media platforms. By effectively distinguishing between genuine and false information in both HGC and AIGC contexts, the proposed model can help mitigate the spread of misinformation, which has significant societal implications. The model's ability to provide consistent authenticity judgments for different versions of the same event addresses a critical need in the era of mixed content, where traditional detection methods often fail due to the sophistication of AI-generated text. Furthermore, the integration of AIGC detection with misinformation detection not only enhances the model's adaptability to emerging scenarios but also provides a more comprehensive approach to information verification. This dual-task framework can be particularly useful for social media platforms, news organizations, and regulatory bodies that aim to maintain information integrity and trustworthiness. Additionally, the detailed analysis and parameter tuning conducted in this study offer practical guidance for future research and development in the field of misinformation detection.

7. Limitations and future works

Despite the contributions of this study, several limitations persist, which warrant further investigation in future research.

- (1) The current approach is limited to investigating the impact of AIGC on textual information. With the rapid advancement and improved capabilities of large models, generated images, audio, and video content are becoming increasingly realistic and prevalent. Therefore, future research should explore authenticity consistency evaluation in scenarios involving multimodal mixtures of AIGC and HGC.
- (2) Due to the inherently opaque nature of most deep learning methods, understanding the mechanisms underlying their superior performance remains a significant challenge. This lack of transparency may raise skepticism regarding the model's predictions. Therefore, improving model interpretability is crucial. For example, integrating dynamically updated knowledge graphs or leveraging the extensive world knowledge of LLMs to provide interpretable explanations for prediction results represents a promising direction for future research.

CRediT authorship contribution statement

Gang Ren: Writing – original draft, Conceptualization. **Li Jiang:** Writing – original draft, Software, Methodology. **Tingting Huang:** Software. **Ying Yang:** Supervision. **Ruida Xie:** Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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